Quantum Optics, IPT5340: Homework #2

Ray-Kuang Lee[1]

[1] *Room 911, Delta Hall, National Tsing Hua University, Hsinchu, Taiwan.*
Tel: +886-3-5742439; E-mail: rklee@ee.nthu.edu.tw[*]

(Dated: Spring, 2026)

Deadline: 11:59PM, Monday, May 4th, 2026.

1. Consider the *Werner state*:

$$\hat{\rho}_{A\otimes B} = \begin{bmatrix} \frac{1-p}{4} & 0 & 0 & 0 \\ 0 & \frac{1+p}{4} & -\frac{p}{2} & 0 \\ 0 & -\frac{p}{2} & \frac{1+p}{4} & 0 \\ 0 & 0 & 0 & \frac{1-p}{4} \end{bmatrix}; \tag{1}$$

with $0 \leq p \leq 1$ and the basis vectors arranged in the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

   (a) Find the *partial transpose* of $\hat{\rho}$ either by taking the transpose of A or B. For example, with respect to the second (B) Hilbert space, the partial transpose is defined as

$$\rho_{im,jn} \to \rho_{in,jm}$$

   (b) Show that the *partial transpose* of $\hat{\rho}_{A\otimes B}$ is only be a physical acceptable state, i.e., only with non-negative eigenvalues, the required range for p is

$$p \leq \frac{1}{3}. \tag{2}$$

   In other words, $\hat{\rho}_{A\otimes B}$ is *inseparable*, i.e., *entangled* when

$$\frac{1}{3} < p \leq 1. \tag{3}$$

2. Plot the Two-mode Squeezed via **QuTip**

   (a) Show that after a 50 : 50 beamsplitter, as shown in Fig. 1, two input Squeezed vacuum states (with orthogonal squeezing angles) are transformed into two Vacuum States.

   (b) However, if you Re-plot the two output jointly, i.e., via (X_1, P_2) and (X_2, P_1), the output Wigner functions are back to the Squeezed ones.

   (c) Or you can Re-plot the two output jointly, i.e., via $(X_1 - X_2, P_1 + P_2)$ as an analogue to the EPR (Einstein-Podolsky-Rosen) pairs.

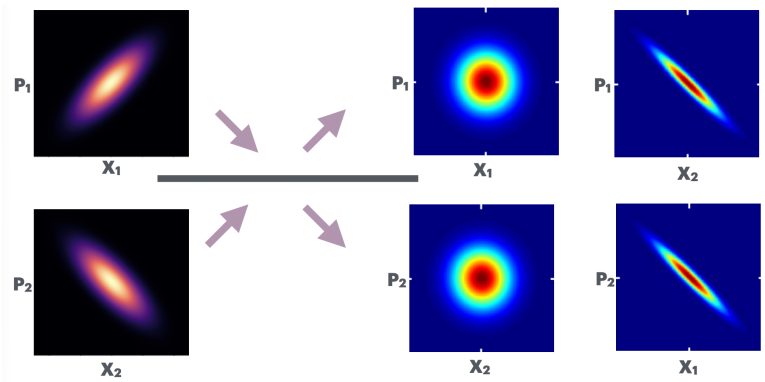

Figure 1: Entangled two-mode squeezed states via a 50 : 50 beamsplitter.

[*] Electronic address: `rklee@ee.nthu.edu.tw`